There is no universal set in Z

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openlogic:set-theory/z:local-snapshot-20261013; CC BY 4.0

Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel thm:NoUniversalSet of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- the Separation axiom schema of Z
- Russell's paradox on the set R = {x : x not in x}

Theorem 0.1. *There is no* universal *set, i.e.,* $\{ x \mid x = x \}$ *does not exist.*

Proof. For reductio, suppose V is a universal set. Then by Separation, $R = \{ x \in V \mid x \notin x \} = \{ x \mid x \notin x \}$ exists, contradicting Russell's Paradox. □

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